

Consilium Mundi

A Game of Interstellar Commerce, Diplomacy and Warfare

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Based on/inspired by the games Diplomacy (rulebook: <http://www.wizards.com/avalonhill/rules/diplomacy.pdf>) and Settlers of Catan (rulebook: http://www.catan.com/en/download/?SoC_rv_Rules_091907.pdf)

Design Objectives:

Consilium Mundi (...and that’s the last time I’m going to write that out in this doc, most likely. I really need a better name, to be honest, as no one will ever be able to remember that as a URL, and “universum” might be a better word than “mundi”, anyway, although the etymology of the whole thing is problematic since, from my understanding, the concept of “outer space” doesn’t really translate. Astra Ad Bellum, Astri Ad Bellum, Ad Astra Bellum...perhaps one of those, whichever is most grammatically correct? But I digress) is a turn-based, multiplayer strategy game. The rules are straightforward and involve very little randomness; in effect, the game will combine the “combat” and negotiation rules of Diplomacy with the “trade rules” of Settlers of Catan in such a way as to create a unique game that *feels* like a board game. In other words, although the game is an iOS, Android, Steam and web-based game, the rules and overall gameplay must be simple enough that players could easily imagine themselves sitting around a table playing a board game version of it (...although yes, there are a few parts of the following rules that would be impossible to duplicate in a board game setting). This, combined with the relative lack of randomness in the rules, should make it abundantly clear to all players that the winner of the game achieved the win *because they were more skilled than everyone else*. Since the turns are simultaneous and the initial game setup is somewhat random luck will of course play a role in winning, but ultimately the game should be one in which players can write up lengthy strategy guides, know the rules by heart and create/memorize specific tactics, and overall form an online community where an individual can become notorious (within the community).

Document Breakdown:

This document is going to be broken down into multiple sections, with new ones added as needed. They’ll go something like this:

- **Section 1 – Game Rules**
Covers the basic rules of play; the specifics of certain game mechanics will be described in other sections, however.
- **Section 2 – Player Communication**
The system I’ve come up with for how players will communicate to each other in-game is unique (and, unfortunately, complicated enough) that it needs its own section to describe.
- **Section 3 – Game Types and Options**
Describes the different game types and options players have when setting up a game – how many players

are involved in a given game, the size of the gameplay area, number of starting home systems, victory conditions, “real-time” turn length, etc.

- **Section 4 – Player Profiles**

What information we track about each of our players, “race creation” and other game-related customization, etc.

Section 1 – Game Rules

Game Overview:

Each player takes on the role of the leader of a space-faring civilization, competing against each other for control of the galaxy. Through diplomacy, espionage, trade, technological advancements and outright war, each player works to take – and maintain – control of as many solar systems as possible.

Game Objective:

Depending on the game type and option, there will be multiple ways that a player can win; these options will be fully described in section 3. Regardless of the “game mode” and other options, however, victory will ultimately boil down to whichever player does the best job of controlling the greatest number of solar systems on the game map.

For the sake of the following rules making sense, however, what would occur during a “standard” game/game setup will be described where applicable. In the case of a standard game, then, a player wins the game if, at the end of a turn, they control more than 50% of all the solar systems on the map.

The Galaxy Map:

The galaxy map is the overall game-play area, and is easy enough to imagine that I’m not going to waste any time drawing it out (yet) – it’s just a bunch of solar systems laid out on a big 2-D(?) map (“the galaxy”), each solar system connected to one or more other solar systems via “wormholes” (think lines or roads...because that’s basically what they are). At the start of the game the map is randomly generated while maintaining as much player balance as possible, the number of solar systems and wormhole connections varying depending on the number of players and the game type/options.

In a standard game there are 8 solar systems per player, plus 1 additional system to prevent a single player from controlling exactly 50% of the systems. Of these, each player begins with 3 home systems (see below). The layout/generation of the galaxy map should be such that every home system connects to 4 other solar systems (i.e. to each other home system, plus 2 additional systems), while every other solar system connects to a minimum of 3 and a maximum 5 solar systems, and any system(s) can be reached from any other system(s) through some combination of system-system connections (i.e. absolutely no “stranded” system(s)). A good logic system for generating these placements and connections will obviously need to be figured out; as a board game it would likely involve some sort of tile system, but as a computer game we can come up with something much better than that.

Solar Systems:

Each solar system provides a certain number of resources to its owner each turn, can contain one or more Starfleets (see below), has 3 or more wormholes that connect it to other neighboring solar systems (through which Starfleet’s travel and attack), and may contain one or more System Upgrades (see below).

Starfleet:

A starfleet represents an indeterminate number of combat-worthy spacecraft grouped together into a single fighting force; a starfleet is the smallest unit of measurement for a collection of spaceships, and can be thought of as a single playing piece in board game terms. A starfleet has a purposefully ambiguous description so as to keep them all equal to one another, despite any imagined differences in the actual spaceship technology used by one space-faring race or another. Thus, every starfleet has the same strength as every other starfleet, and follows the same rules as a starfleet unit controlled by any other player.

Resources:

There are three types of resources in the game: technology and manufacturing (“Tech”), heavy metals and radioactive materials (“Metals”), and organic materials and comestibles (“CHON”; stands for Carbon-Hydrogen-Oxygen-Nitrogen, the basic building blocks of carbon-based life). Every solar system produces zero or more units of Metals and CHON per turn for whichever player controls it; the exact number is randomly generated (while maintaining as much player balance as possible) when the game is first started. Note that only home systems provide Tech resources without the Colony system upgrade (see below for both). Also, as with starfleets, the definition of a resource is ambiguous; for instance, CHON could be native life-forms on a planet or even a large number of “mineable” comets and water-laden asteroids.

In a standard game, non-home solar systems (see “Home Systems” below for home systems) will have a combined resource breakdown of the following; i.e. the following percentages indicate what percent of all non-home systems will provide the indicated resources, evenly distributed around the game board so that the odds of a single player having a bunch of “resource-rich” systems directly adjacent to their home systems is relatively low:

- 15% provide no resources (without system upgrades)
- 25% provide 0 Tech, 0 Metals and 1 CHON
- 25% provide 0 Tech, 1 Metals and 0 CHON
- 20% provide 0 Tech, 1, Metals and 1 CHON
- 5% provide 0 Tech, 1 Metals and 2 CHON
- 5% provide 0 Tech, 2 Metals and 1 CHON
- 5% provide 0 Tech, 2 Metals and 2 CHON

System Upgrades:

System upgrades (or simply, “upgrades”) are purchased with stockpiled resources and provide some benefit to a

single star system. Each upgrade can only be purchased a single time for any given solar system, although any given solar system can have one of each upgrade. Once a system upgrade is built its benefits are available to any player that controls the solar system, regardless of whether or not they were the ones who built it. A player can, however, place orders to have upgrades destroyed should they lose control of a solar system (see below). The following list explains each upgrade available; the rules for purchasing the upgrades, their resource costs, and other related rules are explained later.

- **Starport** – Starports provide two benefits. First, they defend a solar system as if they were a single starfleet. Second, starports are required to supply starfleets (see the “chain of supply” rules below). Unlike other system upgrades, since a starport defends the system as if it were a starfleet, it is automatically destroyed whenever an opposing player takes control of the solar system. The one exception to this rule is if another player has performed a “destabilize system government” act of espionage on the solar system during the same turn (see below).
- **Shipyard** – Players may only build starfleets in a system containing a shipyard; starfleets may not be built at a particular solar system’s shipyard on the same turn in which the shipyard is built there.
- **Colony** – Solar systems with a colony produce 1 additional Tech resource per turn.
- **Mining Facilities** – Any system with mining facilities generates an additional 1 Metals and 1 CHON resource per turn.
- **Wormhole Generator** – Each turn, a single starfleet unit in a solar system containing a wormhole generator may move to any solar system that’s up to 2 systems away, as if that destination solar system and the system containing the wormhole generator were connected via a “normal” wormhole. Only a single starfleet may make this special move on any given turn, and once at the destination system the starfleet must then obey all normal rules regarding attacking, retreating, chain of supply, etc. – in other words the wormhole only exists for the purpose of getting the starfleet to its destination (but will strategically require “defense in depth” by starfleets in systems 2 jumps distance from any enemy wormhole generator).

“Home Systems”:

At the start of the game, each player controls a given number of solar systems – their “home system(s)” – as determined during game setup (see section 3). In a standard game, each home system contains a starport and shipyard, and generates 1 Tech, 1 Metals, and 1 CHON resource per turn. Each player also begins with a single starfleet in each home system. Regardless of game options, only home systems can produce the Tech resource without a system upgrade. Note that the 1 Tech resource produced by the home system is not due to a Colony system upgrade; the player may still purchase a Colony upgrade if they so desire, which would represent complete colonization of the entire solar system.

Also, in almost all game types, the home system(s) of one player may never be directly adjacent to the home system(s) of another player; any wormholes connecting to a home system must connect to either a standard, “un-owned” solar system, or another of the player’s own home systems. On a similar note, every home system for a given player is always connected to every other home system owned by the same player.

How To Play

Game Turns and Phases:

Game play is broken down into a series of turns, each of which is further broken down into several phases. Given the nature of interstellar travel and relativity, a game turn doesn't represent any specific number of days, months, or years. Instead, it represents the amount of time it takes to build enough ships to form a starfleet unit, cobble together a new starport or shipyard, or fly a starfleet from the inner-system planets of a solar system out to a distant wormhole from which they can instantaneously jump to a different part of the galaxy. If it makes you feel any better, think of it as something like 1 to 2 years per turn.

The game turn/phase sequence breaks down as follows, and repeats every turn until someone wins the game:

- 1) **Resource Phase** - Each player collects the resources of each solar system they control
- 2) **Upkeep Phase** –
 - (1) Collected resources are immediately spent to maintain pre-existing starfleets
 - (2) Starfleets that cannot be maintained are automatically removed from game
- 3) **Activity Phase** –
 - Schedule construction of starfleets and system upgrades at designated solar systems
 - Communicate with other players
 - Perform acts of espionage
 - Give orders to your starfleets
 - Schedule the destruction of any unwanted starfleets and/or system upgrades
 - Schedule trade deliveries
- 4) **Resolution Phase** –
 - (1) Resolve Sabotage
 - (2) Destroy starfleets and/or system upgrades scheduled for destruction
 - (3) Starfleet orders are resolved simultaneously
 - (4) Starfleets that lose a battle retreat if possible
 - (5) Any starfleets that end the turn without a chain of supply are destroyed
- 5) **Trade Phase** – Resources are delivered to the appropriate solar system
- 6) **Build Phase** – Starfleets and system upgrades appear at their designated solar system

1. Resource Phase:

At the beginning of each turn, each player receives a lump sum of resources - Tech, Metals and CHON – equal to the cumulative individual resources provided by each solar system they control. For instance, at the start of a game in which each player begins with 3 Home Systems, they would each receive a total of 3 Tech, 3 Metals and 3 CHON.

2. Upkeep Phase:

Immediately after receiving their resources, each player must spend 1 Tech, 1 Metals and 1 CHON resource for every starfleet they have on the galaxy map. This represents the ongoing cost of repairing, refueling and replacing individual ships within the starfleet, recruiting, training, feeding and replacing new and old crewmen, etc. Remember – a game turn represents a year or more of time, during which a given starfleet is doing a lot more than just sitting around waiting to be sent to another solar system. Note that, since this upkeep comes from the player's stores of resources, it may be more or less than the actual number of resources they received on this turn.

The upkeep cost for each starfleet is automatically paid, if possible; if a player didn't wish to maintain what fleets they have, they should have destroyed one or more of them during the Activity Phase. If, however, the Upkeep Phase is reached and the player does not have enough resources to maintain all of their fleets, then one or more fleets will be automatically destroyed. This could happen for any number of reasons, but is most likely to occur when a player has a large number of starfleets and then, during the Resolution Phase of the previous turn, they lose control of one or more solar systems to another player. Use the following process for each starfleet that the player must scrap to determine which specific fleet should be destroyed:

- 1) Count the number of player-owned solar systems between each starfleet and its closest starport, i.e. its "supply chain" as described more fully in Resolution Phase step 5, below. If a single starfleet has more solar systems between it and its nearest "supply-chain" starport than any other starfleets, that starfleet is destroyed. If more than one starfleet ties for farthest distance, *but the only reason for the tie is because all tying starfleets are in the same solar system*, randomly destroy one of them (in game terms there is no reason that one would be more important than the other).
- 2) If more than one starfleet ties for farthest-from-"supply-chain"-starport status in step #1, count the number of solar systems between the supplying solar system of each tying fleet and the Home System closest to that system. Whichever starfleet is both the farthest from a starport and that starport's system is farthest from a Home System is destroyed. Note that the logic for this should also follow a "supply-chain" like link of friendly-only systems; since knowing how close a given starfleet is to a home system is necessary for determining direction of retreating starfleets in some instances, it's something that has to be tracked anyway. As with step 1, if there is a tie due to multiple starfleets being in the same solar system, randomly destroy one of them.
- 3) If there is still a tie, destroy the starfleet that is currently in a solar system with the greatest number of additional starfleets. (In descriptive in-game terms, some of the individual ships from the destroyed starfleet are disassembled for spare parts while the rest are transferred to other starfleets to replace combat losses, etc.)
- 4) If none of the above resolves the situation, destroy a random starfleet that meets the requirements of step 1, 2 and 3.

3. Activity Phase:

This is actually the only phase where players get to make choices of their own; all of the other phases are

resolved automatically based on the decisions each player makes during this phase, as handled by the various rules outlined in each of the other phases (as explained above and below).

The Activity Phase is a lengthy one; its maximum length is determined at the beginning of the game (as part of the game setup/options), measured in hours or even days. The Activity Phase continues until either this time limit is reached and all player turns are automatically ended, or until every player has specifically chosen to “end their turn” (something that – within the computer game – will be visible by all other players; once a player has ended their turn, they can no longer perform any activities, nor will they even receive any communications sent by other players, who themselves will be notified that the player has ended their turn. There should probably be an option to “un-end” your turn, should you decide that you want to change a decision before every other player has ended their turn, although now I digress...).

During the course of this phase, each player has a variety of things they can do, each explained in detail below. Aside from sending communications to other players, each activity is simply a “plan” for what the player intends to do in the Resolution Phase; only messages are actually “done” during this phase. As such, choices made can be undone and redone, changed as diplomatic discussions with other players unfold, etc. There is no particular order to which a player has to perform these various activities; in fact, aside from receiving messages from other players, it is ultimately their choice as to whether or not they even do them at all (...and even when it comes to receiving messages, there’s nothing that says that a player actually has to *read* them).

The following actions can be performed during the Activity Phase:

- **Schedule construction of starfleets and system upgrades at designated solar systems** – The player can take whatever resources they still have and, if they so desire, schedule to spend them on new starfleets and/or system upgrades. Within the limits of their current resources, a player can schedule to build as many starfleets as they want, although they should bear in mind that just because they can build them does not mean that they can afford to pay their upkeep every turn (code/GUI note: there should be some easy way for the player to know how many resources they are gaining a turn versus how many they are spending on starfleet upkeep, and a warning if they decide to build more starfleets than they can support the following turn . . . maybe a running “status” display). By the same token they can schedule to build as many system upgrades as they want, as long as they currently have the available resources to do so and they schedule to build them in a system that does not already have that particular system upgrade. (GUI Note: once a solar system already has a particular system upgrade, the option to build it will either be “grayed out” or a warning message will appear.)

Resource Cost for Scheduled Builds			
	Tech	Metals	CHON
Starfleet	1	1	1
Starport	2	2	2
Shipyard	3	3	1
Colony	0	2	2
Mining Facilities	2	2	1
Wormhole Generator	6	2	0

And to be perfectly clear, scheduling to build a starfleet and/or upgrade is something that’s done *at a particular solar system*; in other words, you schedule what you want to build *and* where you want to build it.

- **Communicate with other players** – In theory the only thing that keeps the Activity Phase going for so long is the back-and-forth communications between players. In a nutshell players send each other messages to make and break alliances, tell each other where they are going to move/attack, discuss the trading of resources, etc. How this all works (and the reason(s) why I don't think it should just be a matter of sending emails/instant messages to each other) will be fully explained in Section 2; for the sake of explaining the game rules, however, it works just to note here that players will be able to communicate with each other in such a way that they can promise each other that they'll perform certain actions, and then decide if they actually want to go through with those actions or instead betray the other player by doing something else instead, etc.
- **Perform acts of espionage** – In addition to building starfleets and system upgrades, a player may spend their Tech resources in a few other ways to sow chaos and discord amongst their enemies. Acts of espionage – depending on what they are – can have an immediate effect, or take effect in the upcoming Resolution Phase. Note that each act of espionage requires 1 Tech resource; this is because Tech is the hardest to acquire resource, and espionage – if performed correctly – can have a major effect on the game.

The number of acts of espionage a player can perform during a given Activity Phase is only limited by their desire to do so and the number of Tech resources they have available to spend.

The different types of espionage and their effect are as follows:

1. **Send Fake Communication** – Send a message to another player, except that the message appears to come from a different player of your choice. Since communications are threaded (albeit restricted to the message's original topic; sorry to be an ass, but you'll have to wait 'til Section 2 for details), any replies to this message will come back to you, your replies to replies will continue to appear as if they were from the other player, and so on.
As this otherwise acts as a standard communication, it is performed completely during the Activity Phase.
2. **Intercept Communication** – Choose two other players; for the entire duration of the Activity phase, you can view every communication exchange between the two of them.
3. **Sabotage** – First, choose a solar system controlled by another player *and adjacent to a solar system that you control*. Then, choose either a single system upgrade or the starfleets present in that system. If you choose a system upgrade, it does not provide its owner with its normal benefits until the player's next Activity Phase; starports stop providing a supply chain and extra defense, shipyards cannot build starfleets (and thus any starfleets scheduled to be built there this turn, aren't), colonies and mining facilities don't produce extra resources, and any starfleet ordered to use a wormhole generator instead just holds position in the system. If you choose to sabotage the starfleets, any movement orders given to them are ignored (although they can still be destroyed by the owning player, and still defend the system if attacked).
4. **Destabilize System Government** – Choose a solar system controlled by another player *and adjacent to a solar system that you control*. During the Resolution Phase of this turn, any orders to destroy System Upgrades in the chosen system are ignored. Additionally, if the system contains a starport and is successfully attacked by another player, the starport is *not* destroyed as per normal combat resolution.

5. **Counter Espionage** – Choose a solar system you control *and that is adjacent to at least one solar system controlled by another player*. During the Resolution Phase, any acts of espionage targeted against that solar system (Sabotage, Destabilize System Government) automatically fail.
- **Give orders to your starfleets** – All starfleet orders are based on their own movement, or the scheduled movement of other starfleets; where do you want them to move to (or should they hold position; the default), who should they provide support to and where should they provide that support (if they aren't moving themselves), and where should they retreat if forced to.

Each starfleet may be ordered to "hop" to any solar system that's connected to the system they are currently in via a wormhole; in other words, each starfleet may move 1 solar system per turn. The exception to this is a wormhole generator; when using the GUI to give orders to a starfleet, if the starfleet is in a solar system with a wormhole generator upgrade that hasn't yet been scheduled for use with another starfleet that turn, the starfleet will be allowed to "choose" from any system within 2 hops of its current location (and once a single starfleet has been ordered to do so, any other starfleets in that system will only be given the option of choosing an adjacent system to move to). In any case, the destination solar system may be one controlled by the player, controlled by another player, or unclaimed by any players; there is no difference in the move type, just in how they are resolved.

A starfleet that has not been ordered to move defaults to a "hold position" order; namely, that they will stay in their current solar system and defend it against any attackers.

Instead of being ordered to physically move to a new solar system, a starfleet can instead be ordered to "support" the orders of starfleet(s) in another solar system. When a starfleet is ordered to provide support, the player must then choose a solar system that has one or more starfleets in it (player-controlled or even those controlled by another enemy), and then select a solar system that's both adjacent to the supporting starfleet *and* adjacent to or the same as the starfleet(s) they are providing support to. If the player wants, a given starfleet can provide support to another starfleet in the same solar system (assuming there is at least 1 other one present) that has been ordered to attack another solar system; in effect, this allows the player to use all of their starfleets in a given system for attacking, without leaving the system defenseless by actually moving all of the starfleets out of it. A player may also order one starfleet to support another starfleet that has been given a "hold position" order rather than a move order. Note that a player may not use a wormhole generator to provide support; starfleets may only move through one. This is to insure that there is always an element of risk involved when using a wormhole generator to attack.

During the resolution phase, the starfleet providing support will act as if it moved to the designated solar system, but only if the fleet it was ordered to provide support to does, in fact, move to that solar system; it will provide additional firepower to the supported starfleet, *regardless of whose side the starfleet is on or who the fleet is attacking*, but it will never physically leave its originating solar system and as such will finish the Resolution Phase in its current system. As explained in further detail below, starfleets ordered to provide support will automatically "cancel" their support and instead defend their starting/current solar system, should that system come under attack during the Resolution Phase. In other words, providing support is a way for a starfleet to help with an attack on another system without leaving its initial system defenseless, and is also the only way in which one player can assist another player in attacking a mutual enemy (since, if both players were to order starfleets to actually move into the same system, they'd end up fighting each other).

Each starfleet can also be given a “rally point”; this is an adjacent, player-controlled solar system that the given starfleet should retreat to should it be attacked by a superior force of enemy starfleets and forced to retreat. If there are multiple starfleets in a single system, each may be given a different rally point if desired. Any starfleet that does not have a rally point set will use the following logic to determine their rally point, should the need arise:

- 1) Determine which player-owned starport or starports (in case there’s a tie) the starfleet is closest to, and what the shortest route is for the starfleet to get to that/those starport(s) through player-owned systems. If there is a single solar system that the starfleet would have to travel through to get to any of those closest starports, that system is the system is the rally point. Note that this follows the “supply-chain” logic outlined in several other locations in this document.
 - 2) If more than one system meets the requirements of step 1, determine the distance to the nearest home system through player-owned systems, again following a “supply-chain”-like logic for determination; the rally point is whichever of the tied systems is the least number of hops from a home system. (i.e. the starfleet will rally towards whichever system is “closest to home”.)
 - 3) If none of the above resolve the situation, the starfleet is randomly assigned one of the solar systems that tied in step 2.
- **Schedule the destruction of any unwanted starfleets and/or system upgrades** – At any time during the Activity Phase, a player may schedule a starfleet or system upgrade to be destroyed. When a starfleet is scheduled for destruction, any orders assigned to it are replaced with a “hold position” order. If a wormhole generator is scheduled for destruction, any starfleets ordered to move through it will instead be ordered to hold position. A shipyard that is currently scheduled to build a starfleet *cannot be marked for destruction until the player un-schedules the starfleet build(s) currently assigned to it*. A starport that is currently the only starport providing a chain of supply to one or more starfleets *cannot be marked for destruction until the player first marks for destruction any starfleets that are relying on it for resupply*. Finally, colonies and mining facilities may be scheduled for destruction without any issues (since nothing really relies on them, at least not in the short term). (GUI Note: clear “reason why cannot perform” messages, possibly w/graphics to indicate problem (e.g. supply chain that will be broken) when invalid user choices made. Particularly important here for large “galaxies”, but obviously nice to have throughout code.)

In any case, if a starfleet or system upgrade that has been scheduled for destruction is then used for a different, contrary purpose (a starfleet is ordered to move, starfleets are assigned to be built at a shipyard, etc.), it immediately is un-scheduled for destruction (i.e. whenever there is a conflict, destruction is automatically trumped). (GUI Note: As with all orders, there will be something to indicate that a given starfleet or system upgrade has been scheduled for destruction.)

- **Schedule trade deliveries** – GUI-speaking, when you’ve selected a solar system that you currently do not own and which borders a system you do own (i.e. is directly connected via a wormhole), you are given the option to schedule a trade delivery to that solar system. If you choose to do so, you can schedule for any (or all) of the resources you still have (haven’t applied to builds or espionage) to be delivered to that solar system during the resolution phase.

Scheduling a trade delivery is literally just that – you’re saying that a certain amount of resources is going to show up at a certain solar system. Excepting that the current owner can’t be you, and that the solar system has to be adjacent to a system you own, the status of whether or not it’s even owned by

another player or which player that is doesn’t matter – whoever owns that system at the end of the Resolution phase is going to be the one receiving the delivery. (Which also means, in theory, that you could end up delivering to yourself if you own the target system at the end of Resolution phase).

4. Resolution Phase:

Once all players have either selected the “end turn” button (or whatever it is), or else the game’s designated maximum Activity Phase time has elapsed, the Resolution Phase begins. This phase is all automatically calculated results based on the various orders and scheduling done by each player during the Activity Phase. Once the calculations are performed and the player’s GUI client is updated (or they re-log back in, etc.), they’ll be able to “view the turn” to get a visual representation of what happened.

The resolution of all orders and scheduling takes place using the following logic; although the end result is viewed as simultaneous, certain orders take precedence over others. For each of the following logical steps, however, all appropriate orders are resolved simultaneously – if any of the following steps could give different results depending on what particular starfleet or solar system the calculation began with, for instance, then there’s a problem with the logic and it will need to be fixed.

- 1) **Resolve Sabotage** – All acts of sabotage that were scheduled during the Activity Phase and which were not based on communication (i.e. Send Fake or Intercept Communication) are resolved. Counter Espionage acts are resolved first, followed by the other applicable acts in any order; multiple acts of espionage may have duplicate results, but shouldn’t otherwise cause any actual conflicting logic.
- 2) **Destroy starfleets and/or system upgrades** – All starfleets and system upgrades that were scheduled for destruction are removed from the galaxy map. Since this takes place before starfleet orders are resolved, these destroyed starfleets and/or system upgrades are ignored (since they no longer exist) for the purpose of resolving the following steps.
- 3) **Starfleet orders are resolved** - This step has several sub-steps. Note that references to “(a starfleet’s) solar system” is a reference to *whichever solar system that starfleet is in at the beginning of the Resolution Phase*, while references to “(a starfleet’s) destination” are a reference to *whatever solar system a starfleet was scheduled to move to, or, if providing support for another unit, the solar system that the support order indicated the starfleet would help the supported unit move to/attack*:
 - a. For each starfleet that was ordered to support another starfleet, check to see if any enemy starfleets were ordered to move (as distinct from a support order) into the supporting starfleet’s current solar system. If at least one enemy starfleet was so ordered, the starfleet that was going to be providing support is instead treated as if it had been given a “hold position” order; it provides no support, does not move anywhere, and instead (tries to) defend its solar system. For all practical purposes, it’s as if the starfleet was actually ordered to hold position (or wasn’t given any orders at all, which would have had the same effect).
 - b. Check each solar system on the map (in any order; it shouldn’t matter) for whether or not one or more starfleets were ordered to move to it. Note that the logic that follows is on a “per system” basis, but for each system must be started at step i and continued sequentially in order to arrive at valid turn results.
 - i. If one or more starfleets owned by a single player were ordered to move to it, and the system is owned by the same player, those starfleets move there.

- ii. If one or more starfleets owned by a single player were ordered to move to it, and there are no other players who have ordered starfleets to move to that system, and that system does not contain any enemy starfleets (regardless of what orders those starfleets were given), then those starfleets successfully do so, are moved to that system, and the system is now considered under control of that player. These starfleets are now considered to be holding position in this new solar system for the purpose of further steps.
- iii. If the system contains one or more starfleets that are ordered to provide support to another starfleet, and the starfleet it is supporting *is not* in the same solar system as the supporting fleet (i.e. not in the system we're currently running logic for), check to see if the solar system it is providing support in (i.e. the supported starfleet's destination system) contains any starfleets owned by another player that were ordered to move to the solar system we're currently running logic for. If so, the supporting starfleet's orders change to "hold position". (In other words, if an enemy starfleet is moving through a wormhole that a supporting starfleet was ordered to "support through", the supporting starfleet gets pushed back out of the wormhole by the moving enemy starfleet.)
- iv. If the system contains one or more starfleets that are ordered to move to another solar system, and that destination system contains starfleets owned by another player that were ordered to move to the system we're currently running logic for, we need to compare the total strength of the two player's "interpenetrating" starfleets to determine if one and/or the other cannot move, and if any supporting starfleets need to change their orders. To do so, add together the total number of starfleets sent from this system to the other system (or vice-versa) with the total number of starfleets that were ordered to support that move *and that originated from the same solar system*. If both sides have the same strength, all starfleets involved change their orders to "hold position". If one side is stronger than the other, all starfleets on the weaker side change their orders to "hold position" while those of the stronger side keep their current orders. (In other words, if two opposing groups of starfleets attempt to use the same wormhole, one or both will be pushed back and unable to take part in any further combat that occurs in the system they were headed to.)
- v. For every starfleet in this system that was ordered to support another starfleet owned by the same player, check to see if the supported starfleet was actually ordered to move to the correct destination system. If it was not (i.e. the player ordered starfleet A to support starfleet B on an attack on System Y, but starfleet B instead was ordered to attack System Z) then the supporting starfleet's orders change to "hold position".
- vi. For every starfleet in this system that was ordered to support a starfleet owned by a *different* player, check to see if the supported starfleet was actually ordered to move to the correct destination system. If it did not, the supporting starfleet's orders change to "hold position". If the supported starfleet was ordered to move to this system, however, a second check needs to be made to see if any of the supporting starfleet's owning player's other starfleets was ordered to move to/attack that system; if they were, then the support order is cancelled and replaced with a "hold position" order. This is to enforce game play logic stating that *no player may support an attack against one of his own starfleets*.

- vii. If a player's starfleet exists in this system, and the system contains one or more starfleets that were ordered to move to a different system (which could, in theory, leave this system "empty", or at the very least with reduced forces in it), and one or more starfleets owned by a different player have been ordered to move to this system, then this system needs to be flagged as "pending" and skipped over until each system has either been marked as "pending" during this step or completed step viii.
- viii. If a player has one or more starfleets in this system, all of which are ordered to hold position and/or the system contains a starport, and the starfleets of one or more players are ordered to move to this system, determine each player's total starfleet strength by adding together the total number of starfleets they sent to that system (or that are already in the system holding position), plus each starfleet that still has orders to support one of those starfleets moving there. Remember that a starport defends a system as if it were a single starfleet. Then, compare each player's total to each other player's total; if a single player's "starfleet strength" is at least 1 greater than the strength of every other player (individually; not the total of all other players together), that player wins the conflict, while all other player's are considered in a standoff. If no single player's starfleet strength is greater than every other's, then the attack move(s) are considered a standoff and there are no winners.

If the winner is the current system owner, or there is a universal standoff, then the current system owner (if any, since this may occur in a currently un-owned system) maintains control of the system, their starfleets remain in place, while the moving/attacking starfleets of every other player return to their original systems and their orders change to hold position. If one of the attacking player's wins, then all other moving/attacking player's starfleets return to their original systems and take hold position actions, the starfleets owned by the winner are moved into the system and the system becomes controlled by that player, and the owner of the starfleets that were previously controlling the system are forced to retreat – they are flagged as "retreating" and, for the time-being, do not count as being present in any solar system for the purposes of resolving starfleet orders (they are handled in the next step of the Resolution Phase).

At this time, if there are any solar systems that have not yet been checked, we go back to step i for that system. If all have been checked at least once, we continue on to step ix.

- ix. Once all systems have been cycled through by steps i through viii, we go back through all "pending" systems to determine what happens there, starting at step i and continuing through the normal logic as stated above. Given those steps, each "pending" system will either no longer be "pending" or will remain in "pending" state. If at least 1 "pending" system is no longer marked as "pending", the system makes note of that for step x, below. .
- x. If the game still thinks there are pending conflicts, *and at least 1 pending system was resolved*, we do another pass of all pending systems, once again using the logic outlined in steps i through viii, followed by a "pending" check in step ix. This continues until we either have no pending systems left, in which case we continue to step xiv, or until we

make a full pass of pending systems without a single pending system being unflagged/resolved, at which time we have to go to step xi.

- xi. When we have gone an entire cycle without a single system being unflagged as "pending", we need to resolve any of the "pending" systems that contain one or more starfleets set to hold position, and one or more starfleets that are set to move to another system, and which have one or more enemy starfleets trying to move into them (i.e. the system is "pending" and also one in which a battle will occur). For each of these systems, we run a "special case" of step viii logic: we perform the logic as normal, except that we count any starfleets that are currently in the system and ordered to move out of it as being "in" the system for the purpose of the total starfleet strength for the owner of that system (i.e. they are treated as having been ordered to hold position for this calculation only).

If, after including these starfleets in the calculation, there is a winner and *it is not the owner of the system*, then we resolve step viii as written, with the winner taking control of the system, their starfleets moving in, all other starfleets involved in the situation having their orders switched to "hold position" (as stated in step viii), and any "hold position" starfleets currently in that system being marked for retreat (again, as stated in step viii). This system is then un-flagged as "pending", allowing our cycle to continue. However, those starfleets that started this Resolution phase in this system and were ordered to move, remain in that state and (at this point in time) do not have their orders changed (they will continue to try and move to their designated system).

If, after performing this calculation, there are either no winners, or the owner of the system would be the winner, we leave the system in "pending" mode and continue on to the next system, beginning at step i again as normal.

If we get to this step and, during the course of it, not a single "pending" system is unflagged as pending following these modified combat conditions, then we go to step xii. Otherwise we go ahead to step xiii.

- xii. For each each "pending with combat" system, we consider them all to be a standoff, deferring to the logic in step viii to indicate which starfleets have their orders changed, which are marked for retreat, etc. As part of this step, some starfleets may be ordered to "hold position" in a solar system that is no longer controlled by their owning player (because that system was "won" by another player during step xi); these starfleets need to be marked for "retreat" (in essence, they took part in the original fight in this system, and, as part of the losing force, must retreat away as with every other starfleet there).
- xiii. If all of the above logic works out correctly, at this stage we should be left with a bunch of starfleets that all want to move into solar systems that are about to be vacated by the starfleets of another player. As such, every starfleet completes its move; it ends up in its destination system, that system comes under the new player's control, and all starfleets that aren't flagged for retreat go to hold position orders.
- xiv. We do a final sanity check of all systems; each system should contain starfleets owned by only a single player or contain no starfleets at all, excepting starfleets that are flagged

for retreat (they don't count). If this is not the case, then we need to throw some sort of error so we can figure out what went wrong and update the logic.

- 4) **Starfleets that lose a battle retreat if possible, or are destroyed** – For each starfleet flagged as retreating, move them to the system that was designated as their rally point during the Activity Phase; if none was designated, use the logic there to determine which one it should be. If the rally point system is owned by the same player as the starfleet, the starfleet successfully moves to that system and is changed to a hold position order. If the rally point system is owned by another player at this point in time, *regardless of whether or not there are starfleets there*, the retreating starfleet is considered to have been destroyed during the fight and is removed from play (i.e. a retreating Starfleet cannot take control of a system owned by another player).
- 5) **Any starfleets that end the turn without a chain of supply are destroyed** – The final step of the Resolution Phase is to check for starfleet chain of supply. To do so, we cycle through every solar system, and perform the following logic:
 - a. If the solar system doesn't contain any starfleets, we skip it.
 - b. If the system contains one or more starfleets, we check if it has a starport system upgrade. If it does, we skip it.
 - c. If the system contains one or more starfleets and it does not have a starport, we check the owner of every solar system that is directly connected to this system via a wormhole. If none of those systems are owned by the player that owns this system (i.e. the system is completely surrounded by enemy systems), then the starfleets in the system are considered cut off from supplies and are destroyed. We can then continue on to the next system.
 - d. If the system is connected to one or more other friendly systems, we check whether or not each of those systems has a starport. If there is at least 1 starport, we're good to go, and we can now skip this system.
 - e. If none of the systems in step d contain a starport, we take note of each of their identities, and then check their wormhole connections; if they connect to another system owned by the player that isn't on our "already checked" list, we continue to trace out their connections until we either determine that the original solar system has no connections to any friendly systems with a starport, or else it does; if it doesn't, then the starfleets don't have a chain of supply, and are destroyed. Otherwise they are fine, and we skip this system and move on to the next one.

(Ultimately there's probably a better programmatic way of doing this...I'm just not sure how . . . keep a running "supply state" for each system with state change recorded during appropriate Resolution phases?)

5. Trade Phase:

Any scheduled trade deliveries arrive at their destination systems. Whoever currently owns that system receives these deliveries, and adds the resources to their current surplus/store of resources.

5. Build Phase:

We cycle through every scheduled starfleet or system upgrade build, and perform the following logic:

1. If the system is no longer owned by the player building the starfleet(s) and/or system upgrades there, the builds are not completed; the player simply loses their resources (or more precisely, they don’t get them back).
2. If the system is still owned by the player, and they are building any starfleets in that system this turn, we check to see if a shipyard is present. If it is, the starfleets are successfully built, and are placed in that solar system. If a shipyard does not exist (which would happen even if it was scheduled to be built here this turn, since – as of this step –it hasn’t been built yet), then the starfleets are “lost”.
3. If the system is still owned by the player, any system upgrades scheduled to be built here, are. Since you can only have 1 of any system upgrade, they are pretty much just flags and so checking for accidentally building multiples at this stage isn’t a big deal (since we’d just be re-toggling the same flag “on”).

At this point in time the turn is complete, and we check to see if any players now meet the requirement(s) to win the game. If so, the game ends and we go to some sort of stat page. Otherwise, the game continues, going to the Resource Phase of the following turn.

Section 2 – Player Communication

During the Activity Phase, players will have an ample amount of time to communicate with each other. In fact, this is an essential part of the game; making and breaking alliances, coordinating attacks, trading resources, etc. Although we could leave this up to regular emails, or even a standardized in-game email-like system, doing so poses several problems:

- We have no control over what players say to each other, and whether or not it’s even game-related. Thus, we have all the legalities that entails, which none of us want to worry about.
- Legal issues aside, we would still have to worry about players being simply abusive to each other, distracting each other with “communication spam” that has nothing positive to provide the game, and other such things that detract from the game play experience and would, in theory, ultimately result in players quitting the game (which, once monetized, is of course not something we want).
- If you allow players to write their own communications, there’s no way that you could fake a communication from one player to another – they would instantly recognize that it’s not written by the same person.

Given these problems, I’ve come up with a “communications system” that will be used by the players as they talk amongst themselves during their plotting and planning. The core of the system starts with a “Diplomacy Interface”, pulled up via a button or something at any point in time during the Activity Phase. Let’s start with that description, and carry on from there:

The Diplomacy Interface

A window displays each player in the game (by their in-game race/civilization name; hereafter referred to as “opponent”, while “player” will refer to the user that’s currently using the interface), how many total message threads there are between the player and the opponent (over the course of the entire game; “Archive Threads”), how many “active threads” there are (from the current Activity phase; this is a subset of the total), and how many unread message threads the player has awaiting from that particular opponent. In addition to

this information, there’s an additional button for each opponent which allows the player to begin a new message thread with them.

Message threads, in game terms, are pretty much synonymous with the idea of an email or “instant messenger” thread – all back-and-forth messages that relate to the same basic topic, as determined by the first thread. Within the context of a given message thread, the specific messages one can send are restricted based on the thread’s topic.

The player can click on the different parts of this interface, which correspond as one would expect to the following sections, detailed below: Archive Threads, the Active Threads, the Unread Threads, or the Start New Thread. Each selection modifies the Diplomacy Interface to display new information, and allow for new interactivity.

Archive Threads:

All messages sent between the player and opponent are listed in a table, each row indicating what the message topic is (it’s “subject line” or summary, basically), how many messages are within that thread, and on what turn each message was sent. This includes messages from the current turn (probably highlighted or something to differentiate them). Clicking on a given message thread will open up the entire thread, and allow the player to respond to that message (regardless of how old it is; for instance, you might be mad that last turn the opponent said they’d trade with you, but they never actually sent the trade goods). See “Responding to Messages”, below.

Active Threads:

This looks and functions the same as the Archive Threads, except that it only shows message threads from the current Activity Phase.

Unread Threads:

This looks and functions the same as the Active Threads, except that it only shows threads that contain one or more messages that have not yet been viewed; as soon as the player opens up any message thread, regardless where they opened it from (from the Archive Thread view, Active Threads, etc.), it is marked as “read” and thus removed from this interface view.

Start New Thread:

This option is different from the others, in that it has the most complicated interface, and is where the core of the communications system resides. Thus, I’m just noting it here, and will explain it fully below under “Sending a New Message”.

Sending a New Message:

When a player chooses to send a new message, they are given a menu of options that indicate the topic of the message they wish to send and the attitude in which they wish it delivered. Message topics cover the various aspects of the game, while message attitude provides the intent of the message (i.e. a threat, promise, plea, etc.), and – after various player selections – will result in a “diplomat” sending the appropriate message to the opponent. That’s right; the player won’t ever actually specify exactly what they’re going to say, merely what message they want to convey and how they want to convey it – the exact details of the wording will be generated by the game and sent to the opponent (just like in real life, rulers of a country have others’ who do much/most of the talking for them).

In most cases, the message topic chosen will require that the user specify a solar system for the message to be about. As such, once the given message topic (as described below) is selected, the user will then have to select

an appropriate solar system on the galaxy map. The interactivity for this will be pretty straightforward: the user selects a message topic that requires specifying a solar system, at which time the diplomacy interface disappears or otherwise moves out of the way, and an alert will appear (probably with the picture of your diplomat – that’ll be awhile in the making, but eventually there will be graphics for that) saying something like (as if the diplomat were saying it) “select which system you wish to discuss (the topic) about”. The player then gets to roam around the galaxy map until they find the system they want; depending on who owns the system, the actual message contents will change. Also, although it’s possible that some messages simply won’t work in reference to particular systems, I think we’ll probably just allow for “crazy-talk” (...and allow a “you’re crazy, that doesn’t make any sense” response). Once the system is selected, a new interface will open up allowing them to choose the message content, the choices depending on what system was chosen. Note that, based on the selected system, an additional, neighboring system may be selectable as well; this will be from a pull-down menu in the “message content” interface rather than a click-on-the-map choice, and will be populated by appropriate solar system names. Once the topic and system are selected, the player then selects an attitude which will change the message content to conform to the desired intent (NOTE: the player should have the option to review the message and change the attitude, prior to actual sending). The possible message topics, attitudes, and resulting intent are described below in order.

Message Topics:

Note that full details of what the actual message’s content is, and what choices are available, would be ridiculous to attempt to write out at this point; the time is better spent during development, when actually coding things. Thus, think of the following message topics descriptions as a general outline of what a given message will be about, not a full explanation of what the content or game/UI-mechanics will be:

- **Attacking a Solar System:** The player is discussing the movement of their own or their opponent’s starfleets into or out of a solar system (i.e. threatens to attack the opponent’s solar system, asks them to move starfleets into their solar system (in theory, because they are going to retreat from it), requests that they attack another opponent’s solar system, etc.). If desired, they can specify from which adjacent solar system they will attack/wish not to be attacked from/think the opponent should launch their attack from.
- **Providing Support:** These messages always require either the currently-selected system to be chosen a second time or else a secondary, adjacent solar system to be selected. The player is discussing using their own or their opponent’s starfleets to support an action involving the selected system (i.e. threatens to support a different opponent’s attack of the system, requests that the opponent support their starfleets moving to this system from the neighboring system, offers to support starfleets moving from the opponent’s system to a neighboring system, etc.).
- **Trade Resources:** The player is discussing a trade delivery to a given system (i.e. threatens to provide resources to a different opponent, offers to send a certain number of resources to the given system and requests that a certain number of resources be sent to one of their systems (or vice-versa), suggests that the opponent send resources to another opponent’s system, etc.).
- **Providing Espionage:** The player is discussing the use of Espionage in the selected system (i.e. threatens to, offers to, or asks the opponent to order one or more of the possible espionage activities to be carried out in a given system).
- **Providing Intelligence:** The player is discussing sharing “communications of interest” concerning another player (i.e. threatens to, offers to, actually does, or asks the opponent to share a copy of an

Archive message from/to the selected player). Unlike most other message topics, this topic does not depend on the user selecting a solar system; instead, one of the required message content choices is the name of the player they are talking about. When actually sending a copy of an Archive message, the player must select one message from their own Archive originally addressed to/from the selected player (i.e. an interface provided subset list of their Archive) which is then copied into this current message, verbatim, along with original message delivery data.

- **Providing Alliance:** The player is discussing a proposed alliance with the recipient of the message. Unlike most other message topics, this message topic does not depend on the user selecting a solar system; instead, one of the required message content choices is a menu of all players in the current game.

Message Attitude:

In all instances of message-sending, a user will be able to select an “attitude” with which to send the message. This will result in different verbiage based on the message topic; the end result is really just a matter of how the final message is randomly generated, although it can completely change the ultimate meaning conveyed by the message; for instance, a “polite” threat to attack an opponent’s solar system would be worded more like a request for that opponent to leave the system “so that hostilities can be avoided”, while an “aggressive” offer for trade would be worded as a threat where the opponent better accept the trade offer “or else”.

Note that, as will be explained in Section 4 – Player Profiles, a given race will have a “default attitude” that will be applied to all messages; thus, a player wishing to play a pacifistic race/civilization might select “polite” as their default attitude, saving them the time of having to select that option all the time.

The following attitudes can be selected for each message sent/received. How they influence a given message’s content is generalized, as the specifics will obviously depend on the message’s topic.

- **Neutral:** This is the default attitude, used to deliver messages in a politically-correct but somewhat indifferent manner appropriate to the context of the message. It may tend towards any of the other attitudes, but only inasmuch as would be appropriate for two diplomats discussing a matter over which neither has any heated feelings.
- **Belligerent:** This is an offensive attitude, used to deliver messages in a less-than politically-correct manner appropriate to the context of the message. It tends towards threats and ultimatums that assume the receiving diplomat has no choice but to acquiesce.
- **Aggressive:** This is a demanding attitude, used to deliver messages in a less-than politically-correct manner appropriate to the context of the message. It tends towards orders, demands, and “you shall” verbiage that assumes the receiving diplomat will comply without argument.
- **Polite:** This is a positive and collaborative attitude, used to deliver messages in a politically-correct and constructive manner appropriate to the context of the message. It tends towards suggestions, requests, deal proposals, and recommendations that assume the best interests of both diplomats will be achieved.

- **Apologetic:** This is a somewhat servile attitude, used to deliver messages in a politically-correct manner appropriate to the context of the message. It tends towards obviously self-serving requests and pleas for assistance from the opponent’s diplomat.

Messages with Attitude:

- **Neutral Attacking a Solar System:** The player suggests the opponent move starfleets into the player’s solar system (in theory, because they are going to retreat from it or otherwise leave it unoccupied) or into another player’s system, or requests that the opponent move out of a solar system which the player believes to be another’s property.
- **Belligerent Attacking a Solar System:** The player threatens to attack the opponent’s solar system or another opponent’s star system. If desired, they can specify what response is required (e.g. “unless you”) from the opponent to avoid implementing the threatened attack.
- **Aggressive Attacking a Solar System:** The player demands that the opponent attack another opponent’s solar system, or emphatically states that they will attack the opponent’s system. If desired, they can specify from which adjacent solar system the attack is to come from.
- **Polite Attacking a Solar System:** The player proposes to move to the opponent’s solar system with their blessing (assumedly because the opponent wishes the player to have control), suggests the opponent move starfleets into the player’s solar system because they are going to retreat from it or leave it unoccupied and wish the opponent to have control, or requests that they attack another opponent’s solar system. If desired, they can specify from which adjacent solar system they recommend the opponent should/not launch their attack.
- **Apologetic Attacking a Solar System:** The player asks the opponent to retreat from the solar system or leave it unoccupied so the player can attack it unmolested, or begs them to attack their or another opponent’s solar system.
- **Neutral Providing Support:** The player suggests that the opponent support their starfleets moving to this system from the neighboring system or that the opponent might want the player to support starfleets moving from the opponent’s system to a neighboring system.
- **Belligerent Providing Support:** The player threatens to provide support to another opponent’s starfleets moving to the opponent’s system from the neighboring system. If desired, they can specify what response is required (e.g. “unless you”) from the opponent to avoid implementing the threatened support.
- **Aggressive Providing Support:** The player demands that the opponent support the player’s, or another opponent’s, starfleets moving to this system from the neighboring system. If desired, they can specify from which adjacent solar system the opponent is to support from.
- **Polite Providing Support:** The player requests that the opponent support their starfleets moving to this system from the neighboring system or offers to support starfleets moving from the opponent’s system to a neighboring system.

- **Apologetic Providing Support:** The player begs the opponent to support their starfleets moving to this system.
- **Neutral Trade Resources:** The player proposes to send a certain number of resources to the system, or suggests that the opponent do so.
- **Belligerent Trade Resources:** The player threatens to send a certain number of resources to another opponent's system. If desired, they can specify what response is required (e.g. "unless you") from the opponent to avoid implementing the threatened trade delivery.
- **Aggressive Trade Resources:** The player demands that the opponent deliver a certain number of resources to the player's system, or orders the opponent to send resources to another opponent.
- **Polite Trade Resources:** The player offers to send a certain number of resources to the given system and requests that a certain number of resources be sent to one of their systems (or vice-versa).
- **Apologetic Trade Resources:** The player asks for a certain number of resources to be delivered to the given system.
- **Neutral Providing Espionage:** The player proposes to order, or suggests that the opponent order, one or more of the possible espionage activities to be carried out in a given system.
- **Belligerent Providing Espionage:** The player threatens to order one or more of the possible espionage activities to be carried out in a given system. If desired, they can specify what response is required (e.g. "unless you") from the opponent to avoid implementing the threatened espionage.
- **Aggressive Providing Espionage:** The player demands that the opponent order one or more of the possible espionage activities to be carried out in a given system.
- **Polite Providing Espionage:** The player offers to order, or recommends that the opponent order one, or more of the possible espionage activities to be carried out in a given system.
- **Apologetic Providing Espionage:** The player asks the opponent to order one or more of the possible espionage activities to be carried out in a given system.
- **Neutral Providing Intelligence:** The player actually shares, or suggests that the opponent share, a copy of an Archive message from/to the specified player.
- **Belligerent Providing Intelligence:** The player threatens to share (assumedly to another opponent) a copy of an Archive message from/to the specified player. If desired, they can specify what response is required (e.g. "unless you") from the opponent to avoid sharing the assumedly incriminating Archive message.
- **Aggressive Providing Intelligence:** The player demands that the opponent share a copy of an Archive message from/to the specified player.
- **Polite Providing Intelligence:** The player offers to, actually does, or requests that the opponent share a copy of an Archive message from/to the specified player.
- **Apologetic Providing Intelligence:** The player asks the opponent to share a copy of an Archive message from/to the current specified player.

- **Neutral Providing Alliance:** The player suggests an alliance with the opponent or states that they will be forming an alliance with another player.
- **Belligerent Providing Alliance:** The player threatens to establish an alliance with the selected player. If desired, they can specify what response is required (e.g. "unless you") from the message recipient to avoid implementing the threatened alliance.
- **Aggressive Providing Alliance:** The player demands specific alliance terms from an opponent.
- **Polite Providing Alliance:** The player offers specific alliance terms to an opponent, in relationship to the message's selected player.
- **Apologetic Providing Alliance:** The player asks an opponent for specific alliance terms.

Responding To Messages:

When responding to a message, the player is shown the text of all previous messages, along with the current message that the opponent sent (again, think email/IM). They are then presented with options for how to respond to the message; this may be a simple reply that doesn't require any additional user interaction, or could be one of the message topics mentioned above. If the latter, the user is shown the galaxy map and asked to select a solar system via the same UI as noted above, except that in this case the message's content will be sent as a specific reply to the thread's topic and thus its verbiage will change to reflect that (e.g. a counter-threat to Attack A Solar System would become a statement of "if you do (message topic), I'll attack your solar system", etc.). Although this may seem like an insane amount of logic to deal with, it will mostly be a matter of pasting together various strings of text based on the current context of the message, and likely not too difficult to pull off.

As with the original message, the player can select an attitude that actually modifies the intent of the response, generating a "response with attitude". The following list governs the initial responses with attitude available for each given message with attitude (listed above in "Sending a New Message") . Further replies will vary depending on this initial response, and will also be touched upon below:

- **Neutral Response With Attitude to:**
 - **Any Neutral Message with Attitude:** The player noncommittally takes the suggestion "under advisement".
- **Belligerent Response With Attitude to:**
 - **Any Neutral or Aggressive Message:** The player rebukes the opponent for their insolence and threatens (specific or unspecific) future actions for any recurrence.
 - **A Belligerent Attacking a Solar System Message:** The player threatens (specific or unspecific) retaliation if the opponent's threat to attack the opponent's solar system or another opponent's star system is carried out, or contemptuously denigrates the opponent's feeble attempt at blackmail.
 - **A Polite Attacking a Solar System Message:** As appropriate to the original message content, the player threatens the opponent with an attack of their system if the opponent goes through with the

proposed attack on the player's solar system, threatens to attack the opponent's solar system whenever they wish to, or rebukes the opponent for their request that the player attack another opponent's solar system.

- **An Apologetic Attacking a Solar System Message:** As appropriate to the original message content, the player threatens to attack the opponent's system if they attack the player's system, or rebukes the opponent's impertinence for their plea that the player attacks another opponent's solar system.
- **A Belligerent Providing Support Message:** The player threatens (specific or unspecific) retaliation if the opponent provides support to another opponent's starfleets moving to the player's system from the neighboring system, or contemptuously denigrates the opponent's feeble attempt at blackmail.
- **A Polite Providing Support Message:** As appropriate to the original message content, the player rebukes the opponent for their insolent request that the player support the opponent's starfleets moving to this system from the neighboring system, or a) rebukes the opponent for the insolent offer of support or b) threatens the opponent if their offer to support the player's starfleets moving to a neighboring system is not fulfilled (not that the player really needs the help, of course).
- **An Apologetic Providing Support Message:** The player rebukes the opponent's impertinence for their plea that the player supports the opponent's starfleets moving to this system and if desired may threaten the opponent if ever such a plea is again sent.
- **A Belligerent Trade Resources Message:** The player threatens the opponent with retaliation (specific or unspecific) if they send the specified resources to another opponent's system, or contemptuously denigrates the opponent's feeble attempt at blackmail.
- **A Polite Trade Resources Message:** The player threatens (specific or unspecific) the opponent if their offer to send a certain number of resources to the given system is not fulfilled, and rebukes the opponent for the insolent request for resources from the player.
- **An Apologetic Trade Resources Message:** The player rebukes the opponent's impertinence for their plea for a certain number of resources to be delivered to the given system and if desired may threaten the opponent if ever such a plea is again sent.
- **A Belligerent Providing Espionage Message:** The player threatens the opponent with retaliation (specific or unspecific) if any espionage activity performed in the given system.
- **A Polite Providing Espionage Message:** As appropriate to the original message content, the player threatens (specific or unspecific) the opponent if their offered espionage is/not carried out in the given system, or rebukes the opponent for their insolence in begging for espionage activities by the player.
- **An Apologetic Providing Espionage Message:** The player rebukes the opponent's impertinence in begging for espionage activities by the player and if desired may threaten the opponent if ever such a plea is again sent.
- **A Belligerent Providing Intelligence Message:** The player threatens the opponent with retaliation (specific or unspecific) if the opponent carries out their threat to share (assumedly with another opponent) a copy of an Archive message from/to the current owner of the specified system, or contemptuously denigrates the opponent's feeble attempt at blackmail.

- **A Polite Providing Intelligence Message:** As appropriate to the original message content, the player threatens (specific or unspecific) the opponent if the offered message is not shared, or rebukes the opponent for their insolence in begging the player for a copy of an Archive message from/to the current owner of the specified system.
- **An Apologetic Providing Intelligence Message:** The player rebukes the opponent's impertinence in begging for any intelligence (which they would be incapable of understanding anyway) from the player and if desired may threaten the opponent if ever such a plea is again sent.
- **A Belligerent Providing Alliance Message:** The player threatens the opponent with retaliation (specific or unspecific) if the opponent carries out their threat to establish an alliance with the current owning player of the selected system, or contemptuously denigrates the opponent's feeble attempt at blackmail.
- **A Polite Providing Alliance Message:** The player threatens (specific or unspecific) the opponent if their offered specific alliance terms are/not implemented in relationship to the current owning player of the selected system, or rebukes the opponent for their insolence in begging the player for alliance terms.
- **An Apologetic Providing Alliance Message:** The player threatens (specific or unspecific) the opponent if their own specific alliance terms are not accepted by the opponent, or rebukes the opponent for their contemptuously pitiable attempt to subordinate themselves to the player.
- **Aggressive Response With Attitude to:**
 - **Any Neutral, Aggressive, Polite or Apologetic Message:** The player can reject the opponent's suggestion/demand/request/plea "out of hand", or agree "for their own best interest". If desired, they can also demand that the opponent perform a specific activity in exchange for the player's agreement.
 - **Any Belligerent Message:** The player can order the opponent to do no such thing and/or denigrate the threat and/or demand specific compensation (e.g. resources) from the opponent for wasting their time.
- **Polite Response With Attitude to:**
 - **Any Neutral Message:** The player can agree or decline to carry out the opponent's suggestion, with thanks for their consideration.
 - **A Belligerent Attacking a Solar System Message:** As appropriate to the original message content, the player can promise to vacate the threatened solar system or promise not to interfere with the opponent's attack on another opponent's star system, or regret that the opponent's message could not be understood. If the opponent specified a requirement that must be met to avoid their threat, the player can promise to meet the requirement, or offer an alternative to the requirement.
 - **Any Aggressive Message:** The player can either regretfully agree or disagree to the opponent's demands.
 - **Any Polite or Apologetic Message:** The player can accept or reject the opponent's proposals/pleas "with all due consideration". If desired, they can also provide counter-proposals to the opponent.

- **A Belligerent Providing Support Message:** The player can promise to vacate the threatened solar system, or regret that the opponent’s message could not be understood. If the opponent specified a requirement that must be met to avoid their threat, the player can promise to meet the requirement, or offer an alternative to the requirement.
- **A Belligerent Trade Resources Message:** If the opponent specified a requirement that must be met to avoid their threat, the player can promise to meet the requirement, or offer an alternative to the requirement.
- **A Belligerent Providing Espionage Message:** If the opponent specified a requirement that must be met to avoid their threat, the player can promise to meet the requirement, or offer an alternative to the requirement.
- **A Belligerent Providing Intelligence Message:** If the opponent specified a requirement that must be met to avoid their threat, the player can promise to meet the requirement, or offer an alternative to the requirement.
- **A Belligerent Providing Alliance Message:** If the opponent specified a requirement that must be met to avoid their threat, the player can promise to meet the requirement, or offer an alternative to the requirement.
- **Apologetic Response With Attitude to:**
 - **Any Message:** The player can vociferously agree to comply with whatever the opponent desires (while abjectly apologizing for their own stupidity in not already having done so), or proclaim undying loyalty to the opponent while abjectly admitting that what the opponent desires cannot be accomplished.

Section 3 – Game Types and Options

Before a game can be played it must, of course, be started. In the case of this game, it is up to an individual player to set up a game, “post” it to the server, and await other players to join so that the game can be played.

Game Interface:

From the initial screen/interface that a player reaches after logging in (see Section 4 for more information on user accounts), they will be provided with a list of games they are already playing in (Currently Active Games) – which will include basic data about the game such as what type it is (along with an ID of some sort), what turn it’s on, time remaining until end of the current turn, and (likely, as I assume it’d be helpful) whether or not there are any pending messages awaiting the player in that particular game. Aside from that list there will be two additional buttons; a “Join Game” button which will display a list of “pending” (i.e. not yet started) games that other players have set up via the second button, and the “Start New Game” button:

Currently Active Games:

Clicking on a currently-active game will bring up another window showing the full game details – a list of what game options are being used, what turn it is, what players/civilizations are involved and how many systems and starfleets they each own (unless Fog of War and/or Uncharted Territory game options are enabled; see below),

what messages are awaiting you from each player, etc. They then have a Close button (to exit back to the Game Interface), a Play Game button (which will take them in to this game), or a Quit Game button (which, if clicked, will pop up a “are you sure?” alert and, if selected, will remove this game from the player’s list of active games, marking the player as having lost this game in the process, but as far as the game itself is concerned their starfleets and systems will remain in play and simply follow default “don’t do anything” behavior from then-on out; other players will be notified that this player’s “government has collapsed”).

Join Game:

If the player decides to join a pre-existing game, they will be shown a list of all that are available (as mentioned above) and can then click on one to see that game’s details, as with their list of currently active games. They obviously won’t have a Quit Game button, however, and if they select Play Game they will be given a menu of all of their civilizations (see Section 4 for details), and after selecting which civilization to play the game as the server will check to see if the game has begun yet and, if not (and assuming there’s still player slots available) the user will be added to the game; if the game has been “filled” during this time, however, the user will be notified and taken back to the Join Game screen (which will be updated by the server to reflect the current state of the different available games as of this new point in time).

Start New Game:

This interface has a primary selection area for “game type”, as well as multiple sub-sections for various game options. When first entering this mode, the game type will be set to “Standard” and the options all filled out appropriate for that game mode. By selecting a different game mode, the various options will be auto-populated with the default game options for that game type. If a user customizes any of the game options, the game type will automatically switch to “Custom”; selecting “Custom” from the list of game types will initially display the options for a “Standard” game type. (i.e. if you select a Custom game, then you start with the options of a standard game, and customize from there; if you start messing with game options and are not currently set to a Custom game type, then the type switches to Custom with the options staying as they currently are).

There will also be a selection for which “civilization” the player wishes to play during this game. See Section 4 for details on creating civilizations.

What follows is a list of all the game options, describing how they affect the game and what choices the player can make in selecting them. After that each of the game types is described, indicating what their default settings are. Note that, unless noted otherwise, “Standard” for a category refers to those game values described in Section 1. (UI Note: There will be some sort of “hover tool-tip”, or other Help system in place, to explain some of these options)

Game Options:

- **Players:** The range of players involved in the game. Once the minimum number is met, the server will wait for some pre-defined additional time (5 minutes?) to see if any more can be “slotted” into the game to reach the upper end of the player count before it begins.
Selections are:
 - **3 to 5**
 - **4 to 6**
 - **5 to 8** – Note that this is the “standard” value for most games.
 - **6 to 10**

- **System Density:** Defines how many total systems will be generated on the galaxy map. Note that this affects total number of systems only; home systems are included in these numbers. Selections are:
 - **Sparse** – 4 systems per player, plus 1 additional.
 - **Light** – 6 systems per player, plus 1 additional.
 - **Standard** – 8 systems per player, plus 1 additional.
 - **Heavy** – 10 systems per player, plus 1 additional.
- **Home System Density:** Defines how many home systems each player begins with. Selections are:
 - **Sparse** – 1 home system per player.
 - **Light** – 2 home systems per player.
 - **Standard** – 3 home systems per player.
 - **Heavy** – 4 home systems per player.
- **System Resources:** As explained in Section 1, there’s a randomized disbursement of resources in the (non-home) solar systems. This option sets those frequencies and values (note that “Standard” is – as expected – the percentage distribution listed in Section 1. It is repeated here for clarity). Selections are:
 - **Barren** – 50% provide no resources (without system upgrades); 15% provide 0 Tech, 0 Metals and 1 CHON; 15% provide 0 Tech, 1 Metals and 0 CHON; 10% provide 0 Tech, 1, Metals and 1 CHON; 5% provide 0 Tech, 1 Metals and 2 CHON; 5% provide 0 Tech, 2 Metals and 1 CHON.
 - **Sparse** – 30% provide no resources (without system upgrades); 20% provide 0 Tech, 0 Metals and 1 CHON; 20% provide 0 Tech, 1 Metals and 0 CHON; 20% provide 0 Tech, 1, Metals and 1 CHON; 5% provide 0 Tech, 1 Metals and 2 CHON; 5% provide 0 Tech, 2 Metals and 1 CHON.
 - **Standard** – 15% provide no resources (without system upgrades); 25% provide 0 Tech, 0 Metals and 1 CHON; 25% provide 0 Tech, 1 Metals and 0 CHON; 20% provide 0 Tech, 1, Metals and 1 CHON; 5% provide 0 Tech, 1 Metals and 2 CHON; 5% provide 0 Tech, 2 Metals and 1 CHON; 5% provide 0 Tech, 2 Metals and 2 CHON.
 - **Dense** - 20% provide 0 Tech, 0 Metals and 1 CHON; 20% provide 0 Tech, 1 Metals and 0 CHON; 20% provide 0 Tech, 1, Metals and 1 CHON; 15% provide 0 Tech, 1 Metals and 2 CHON; 15% provide 0 Tech, 2 Metals and 1 CHON; 10% provide 0 Tech, 2 Metals and 2 CHON.
- **Home System Development:** Sets the number of resources generated by each home system, as well as what system upgrades and number of starfleets are present at the beginning of the game. Note that the listed resources produced do not include those provided by system upgrades; they are

the resources naturally produced by the system alone. Regardless of the Home System Development selection for a given game, each player starts with +1 Tech, +1 Metals and +1 CHON for each Starfleet they start with.

Selections are:

- **Colony Worlds** - Each home system produces 0 Tech, 1 Metals and 1 CHON resources, contains a single starfleet, and has colony and starport system upgrades.
 - **Standard** – Each home system produces 1 Tech, 1 Metals and 1 CHON resources, contains a single starfleet, and has colony, shipyard and starport system upgrades.
 - **Advanced** - Each home system produces 2 Tech, 1 Metals and 1 CHON resources, contains a single starfleet, and has colony, shipyard, starport, and mining facilities system upgrades.
 - **Expanding Civilization** - Each home system produces 2 Tech, 1 Metals and 1 CHON resources, contains 2 starfleets, and has colony, shipyard, starport, and mining facilities system upgrades.
 - **Fading Empire** - Each home system produces 3 Tech, 0 Metals and 0 CHON resources, contains 1 starfleet, and has colony, shipyard, starport, mining facilities and wormhole generator system upgrades.
- **Victory Conditions:** Specifies what has to happen for a given player to win the game. Should a game ever reach a state in which the victory conditions cannot be met, the game results in a tie (possibly between only a certain number of the players, rather than all) and ends. All victory conditions are tested for at the end of a game turn, or the end of the game should the Turn Limit game option (see below) be set to anything other than “No Duration”.
Selections are:

- **Standard** – Once a single player controls over 50% of all solar systems, they win. If the game ends before this requirement is met, the player who owns the most solar systems wins (or multiple players tie, if more than one owns the same, greatest number of systems).
- **Galactic Domination** – A player wins if they control more than 50% of all the home systems in the game and no other player controls more than 30% of all home systems. If the game ends before these conditions are met, if a player controls more than 50% of all the home systems they win; if no player controls more than 50% of all home systems, but multiple players control more than 30% of all systems, those players tie. The game automatically ends in a tie if only two players control any solar systems (i.e. all but 2 players have been “destroyed”), the victory conditions have not been met, and neither remaining player controls more than 60% of *all* solar systems. (It is unlikely anyone would ever finish one of these games if they reached a state where there were only two players left and they both controlled half of the map)
- **Last Civilization Standing** – A player wins if they are the only player left in the game that owns at least one solar system containing a starport. If the game ends before this requirement is met, the player owning the greatest number of solar systems with the starport upgrade wins (or multiple players tie, if more than one owns the same, greatest number of starport-containing systems). (Design Note: The reasoning here is that if no other players have any starports, then no other player has a chain of supply, and thus at the

end of the turn every starfleet not owned by the winning player would be destroyed. And, although that doesn’t rule out someone else winning, it basically gives the winning player a free turn to conquer as many solar systems as possible which, ultimately, *will* likely result in them winning.)

- **Corporate Takeover** – A player wins when they control solar systems containing Starport and Mining Facilities system upgrades that number more than 30% of the total number of solar systems in the game. If the game ends before these conditions are met, the player that owns the greatest number of systems containing both a starport and mining facilities wins (or multiple players tie, if they each own the same number of qualifying solar systems). (Design Note: No matter what game options a player selects, there shouldn’t be any way to start the game meeting these victory conditions; the closest you could get is 1 system away from “over 30%”.)
- **Gunship Diplomacy** – A player wins if they own a number of starfleets equal to or greater than the total number of solar systems in the game. If the game ends before this requirement is met, the player with the greatest number of starfleets wins (or multiple players tie, if more than one owns the same, greatest number of starfleets).
- **Turn Duration:** This option specifies the maximum amount of time a given turn of play can last for. The timer begins from the moment that a turn begins (or, in the case of a new game, from the moment that the server has finished generating the game and assigning all players to it, thus “beginning” it) and lasts until either the given duration elapses (measured in real-time; not based on how long a given player is actively playing the game) or until all players have clicked the End Turn button. (Design Note: Although it’s been mentioned before there isn’t actually any part of this design doc that specifies where an “End Turn” button would appear. It’ll be somewhere in the UI...I just don’t know where, exactly)
Selections are:
 - **1 Hour**
 - **2 Hours**
 - **4 Hours**
 - **8 Hours**
 - **16 Hours**
 - **24 Hours** – This is the default/standard duration.
 - **36 Hours**
 - **2 Days**
 - **3 Days**
 - **No Duration** – The game continues until all players have clicked the End Turn button. (Design Note: Might not implement this as there’s a very good chance of games never “finishing”)

- **Turn Limit:** The maximum number of turns that a game can go for before the game automatically ends. In most cases this will result in all remaining players tying. (See Victory Conditions, above, for exceptions)
Selections are:
 - **15 Turns**
 - **24 Turns**
 - **40 Turns**
 - **60 Turns**
 - **100 Turns**
 - **No Limit** – This is the default/standard.
- **No Weekend Turn Duration:** This is a toggle on/off game option. If on, during the course of the weekend (starting at 12:00AM Saturday morning and ending 11:59PM Sunday evening, server time) any game with the Turn Duration option set to anything other than “No Duration” ignores the duration setting; code-wise, the “turn timer” should be paused the moment the weekend begins, and unpaused the moment it ends. A turn can still end if all players hit the “End Turn” button, however. Default setting is on. (My assumption is that players will prefer not to have to worry about a turn-based game over the weekend; I imagine this to be more of a “waste time at work” kind of game than anything else. And there’ll be at least one “weekend warrior” game type, below, regardless)
- **Disable Communications:** This is a toggle on/off game option. If on, the Diplomacy Interface is disabled and no Espionage acts requiring communications are allowed (Send Fake Communication, Intercept Communication). The default is off.
- **Fog of War:** This is a toggle on/off game option. If on, any solar systems that are not adjacent to a solar system owned by the player are “in the fog”; these systems display their (non-system upgrade generated) resources and their owner, but provide no details as to what system upgrades are present or how many starfleets are in the system. The default is off. (Design note: To prevent the wormhole generator from being even more powerful than it already is, that system upgrade has no effect on the fog of war; a player can still move to an unknown system, but they must guess as to its contents)
- **Uncharted Territory:** This is a toggle on/off game option. If on, each player begins the game only being able to “see” their home systems, as well as each system connected to their home systems via a wormhole; in other words, any solar system that is not within 1 jump of one of their home systems begins the game invisible. As players conquer new systems, adjacent (connected via a wormhole) systems become visible on the galaxy map. Additionally, if this option is on, other players are likewise “invisible” to the player until they conquer a system adjacent to a system controlled by that player; until this happens, these “invisible” players cannot be communicated with via the Diplomacy Interface, they cannot be referenced in communications to other players (although it’s possible one player could send a message to another that includes references to a player that’s still “undiscovered” by the recipient), and Acts of Espionage cannot be targeted against them. Once a system or player is “discovered” during the game, they remain so throughout the remainder of the

game (although system data may be constrained if Fog of War is “on”). (UI Note: All game-support player lists, displays, etc. must conform to map “invisibility” of undiscovered systems/players.)

- **Scorched Earth:** This is a toggle on/off game option. If on, whenever a player takes control of a solar system from another player, all system upgrades are automatically destroyed (there is no way to prevent this). The default is off.
- **Alien Artifacts:** This is a toggle on/off game option. 10% of all non-home solar systems, randomly chosen at the start of the game, provide an additional 1 Tech resource. The default is off.
- **Private Game:** This is a toggle on/off game option. If on, the game can only be joined by other players who are on the user’s “friend list” (see Section 4, below, for details); the server will try to randomly assign players from the friend list who wish to join the same type of game, as usual, but the server will not go to the “public list” to fill the player slots, nor will it allow other, non-friend players to see the game in a list of available games to join. The default is off; as this option is also never on except during a custom game, it is not mentioned below.

Game Types:

The following “pre-designed” types of games can be selected for play. As noted above under Game Options, selecting one of these game types will set the various game options to a specific value (as explained below), while changing the options of a game type will automatically switch that game type to “Custom”. Every game type will have a description that goes along with it (the description is listed below – subject to change, of course – following the game type’s name).

Standard: The first Great Interstellar Survey is complete; Wormholes have been discovered, their pathways mapped, new solar systems and resources revealed... and you're not alone! With alien contact has come friction, and whether you want to call it the Great Game, manifest destiny, or simply survival of the fittest, this corner of the galaxy just isn't big enough...

- **Players** – 5 to 8
- **System Density** – Standard
- **Home System Density** – Standard
- **System Resources** – Standard
- **Home System Development** – Standard
- **Victory Conditions** – Standard
- **Turn Duration** – 24 Hours
- **Turn Limit** – No Limit
- **No Weekend Turn Duration** – On
- **Disable Communications** – Off

- **Fog of War – Off**
- **Uncharted Territory - Off**
- **Scorched Earth – Off**
- **Alien Artifacts - Off**
- **Plan Ragnarok:** First Contact occurred in a brilliant flash of thermonuclear weapons and vaporized metal, and all subsequent diplomatic attempts have failed. Battle plans have been drawn and starfleets readied; it's every civilization for itself, and you'll either wipe the galaxy clean of all alien scum or watch your race driven to extinction in the process.
 - **Players – 4 to 6**
 - **System Density – Light**
 - **Home System Density – Standard**
 - **System Resources – Standard**
 - **Home System Development – Advanced**
 - **Victory Conditions – Last Civilization Standing**
 - **Turn Duration – 24 Hours**
 - **Turn Limit – 40 Turns**
 - **No Weekend Turn Duration – On**
 - **Disable Communications – On**
 - **Fog of War – On**
 - **Uncharted Territory - Off**
 - **Scorched Earth – On**
 - **Alien Artifacts – Off**
- **Day of Reckoning:** Prepare for 24 hours of quick thinking, backstabbing, and destruction.
 - **Players – 3 to 5**
 - **System Density – Light**
 - **Home System Density – Standard**
 - **System Resources – Standard**
 - **Home System Development – Expanding Civilization**
 - **Victory Conditions – Galactic Domination**

- **Turn Duration** – 1 Hour
 - **Turn Limit** – 24 Turns
 - **No Weekend Turn Duration** – Off
 - **Disable Communications** – Off
 - **Fog of War** – Off
 - **Uncharted Territory** - Off
 - **Scorched Earth** – Off
 - **Alien Artifacts** – Off
- **Weekend Warrior:** Stock up on snacks, energy drinks and coffee, because you’re not going to get any sleep for the next two days.
 - **Players** – 4 to 6
 - **System Density** – Standard
 - **Home System Density** – Standard
 - **System Resources** – Standard
 - **Home System Development** – Expanding Civilization
 - **Victory Conditions** – Standard
 - **Turn Duration** – 2 Hours
 - **Turn Limit** – 40 Turns
 - **No Weekend Turn Duration** – Off
 - **Disable Communications** – Off
 - **Fog of War** – Off
 - **Uncharted Territory** - Off
 - **Scorched Earth** – Off
 - **Alien Artifacts** – Off
- **Trade Wars:** A new sector of the galaxy has been discovered, rich with minerals and metals to fuel your ever-expanding galactic empire. Unfortunately you’re not the only one out there; the race is on to claim these virgin lands for the glory of your people...and the size of your galactic bank account!
 - **Players** – 5 to 8

- **System Density** – Heavy
- **Home System Density** – Light
- **System Resources** – Dense
- **Home System Development** – Colony Worlds
- **Victory Conditions** – Corporate Takeover
- **Turn Duration** – 24 Hours
- **Turn Limit** – 60 Turns
- **No Weekend Turn Duration** – On
- **Disable Communications** – Off
- **Fog of War** – Off
- **Uncharted Territory** - Off
- **Scorched Earth** – Off
- **Alien Artifacts** - Off
- **Post-Apocalypse:** Following the end of a devastating interstellar war, the surviving civilizations struggle to leave the ashes of their home worlds and lay claim to the now-barren galaxy, ensuring that the destruction never again threatens their race by subjugating their foes through an overwhelming show of galactic force.
 - **Players** – 4 to 6
 - **System Density** – Light
 - **Home System Density** – Sparse
 - **System Resources** – Sparse
 - **Home System Development** – Fading Empire
 - **Victory Conditions** – Gunship Diplomacy
 - **Turn Duration** – 24 Hours
 - **Turn Limit** – 100 Turns
 - **No Weekend Turn Duration** – On
 - **Disable Communications** – Off
 - **Fog of War** – On
 - **Uncharted Territory** - On

- **Scorched Earth** – Off
- **Alien Artifacts** – On

Section 4 – Player Profiles

Given the slow pace of this game and its “strategic board game”-like nature, it is assumed that a given player will be playing multiple games simultaneously and, given the nature of online communities, will want to be able to view their personal record as well as that of others’. Although some aspects of the online gaming experience will be handled outside of the game – such as in a website forum, for instance – others will be viewable from within the game itself.

What follows are descriptions of various parts of the “game experience” that occur within the client program but outside of a game. Although I’m not going to go into details as to where all of these options reside, in general there will be some sort of “main menu” – same place where you can start a new game – which you get to after logging in.

User Accounts and Logging In:

Before a player can do anything, they need to log in. The initial login screen will have a location for their user name, password, a “forgot password or user name?” button, and a “create account” button. Other than logging in, all of the other options will take place on a web page – it’s easier to handle account creation there (since we can use a “type the word you see in the picture” security measure to stop/help with account abuse, amongst other reasons). When a user has logged in, they’ll go to the Main Menu.

User Account Details:

Here’s the information a given person will need to provide when creating an account. We will use generic security measures such as email verification to try and keep account abuse down to a minimum.

- **Password** – Minimum-character restriction (6?) but beyond that I’m not really concerned about “tough passwords”; if someone wants to use a stupidly-simple password, let them.
- **Email Address** – Valid email address, and we will reject multiple user accounts using the same address.

Changing Account Information:

As long as account creation is handled through a web page, so should changing account details. If the account systems get moved inside the client, then changing these details will have to be incorporated into the client (of course).

Player Profile:

Once a user account is created and the player has logged in, they will be able to get to a Player Profile interface (or, perhaps, will be forced to one the first time they log in). Here they can create “civilizations” that will be used when playing different games, and change any player/user account details that are controlled from within the client. As part of this, all of their premade civilizations will be visible.

- **Display Name:** This must be a character set unique to the game instance. Logic for testing this will be straightforward enough; whitespaces will be removed from the display name and it will be converted to lowercase, and as long as the result doesn't match any other Display Name already stored for the particular game then it's an acceptable display name.
- **Civilization List:** Each civilization the player has made will be listed here, showing its picture, name, and all the other details associated with that civilization (see below). Next to each civilization is a Delete button; clicking this will bring up an alert verifying the user wants to delete this civilization, and if the player verifies deletion it's removed from the list. Note that since a deleted civilization might be actively involved in a game it isn't actually deleted out of the database; it just doesn't appear in the player's civilization list and can't be selected when joining new games. The player will continue to “be” this civilization in any currently-playing games, however.

Also note that a civilization cannot be edited; once it has been created it can be deleted and “remade”, but cannot be changed as that could be abused by a player fairly easily (by, say, switching your civilization's name to match that of another player you're currently involved in a game with, with the intent of confusing other players in that game). Note that, from a database standpoint, we need to save civilizations on a per-player basis, each with their own unique ID so that a player (or different players) can create civilizations that have the same names without their being any conflict. Although this too could result in confusing games, at least every player would be aware from the get-go that there were multiple players with the same/similar-sounding names...and ultimately it would be unfair to prevent users from creating similar/same-sounding civilization names, as not everyone is going to be able to come up with non-human-sounding alien races.

- **Create New Civilization:** The player can set up a new civilization to play in games. See the next section for details.

Creating a New Civilization:

The following list of options have to be filled out to create a new civilization. Each will have a “help tooltip” of some sort to explain what the field represents, as well as providing examples. Additionally, there will be one or more “communication examples” on the interface that will demonstrate how the various choices will be used in-game; these examples will appear at the end of this list:

- **Civilization Name:** The display name for this civilization, such as “Terran Empire” or “Alpha Centauri”. There will also be a checkbox for whether or not the civilization name should be referred to as “the” (“Terran Empire” would be, for example). This name will be checked for “bad words”.
- **Racial Name:** The name for the alien race (or, possibly, races) that populate your civilization; these should be plural. Examples are “humans”, “alpha centaurians”, “varied amphibian bipeds”, etc. Note that these won't always be used in communications – sometimes non-species-specific words will be used instead (see examples, below).
- **Ruler Title:** The title by which the civilization's ruler – i.e., the player – will be referred to as, such as “President” or “Grand Pooh-Bah”.
- **Standard Attitude:** Contains a pull-down list of all the different message attitudes, as explained in Section 4. All messages sent by a player actively playing this civilization will default to this attitude type.

- **Flag and Colors:** The player creates their flag by selecting 3 different colors (which could, in theory, all be the same) and two images, one of which appears directly over the solid-color background and the second of which appears over both the background and the first image. Both images can be scaled and repositioned within this area to further aid in customization. By combining images of simple shapes for both images, a simple shape plus a more complex image, or two complex images, each of which is assigned one of the 3 selected colors (first color is for solid background, second applies to the first image, and the third applies to the second image), a large selection of flags should be possible to create. The flag and colors will be used in various ways in the game: as part of messages to other players, for coloration of starfleet and/or system upgrade icons, “sphere of control” shading on the galaxy, map, etc. (Implementation note: this is one of those pieces of the game that will evolve/improve over time, in theory eventually including some sort of “flag submission” system where users can send in flag images formatted a certain way that can then be added into the game.)
- **Diplomat’s Image:** The player selects from various pictures that represent the diplomat(s) who communicate with other players. These images will probably be a two-choice selection, one for the background of the image (the flag being one – and probably the default – option) and one for the “photograph” of the diplomat(s). These will be drawings (or possibly even Photoshop-altered photos) of the actual diplomat (who could be human, an alien...pretty much anything we put in there), a computer screen or robot, etc. – basically, anything that a player might want to have appear in a message sent to another player can be here. In addition to the images themselves, there will be some sort of color-tint system to allow for further customization. (Implementation note: like the flags, this is going to be developed over time and will eventually probably have some sort of user-uploaded system as well.)
- **Communication Examples:** Two semi-random sentences should be constructed based on the civilization choices to help determine whether or not the civilization name and ruler title “work”. These will probably be tailored to go along with the Standard Attitude setting as well, but for now let’s just use these: “(ruler title) (user display name) of (the) (civilization name) requests an alliance between the people of our two great civilizations.” “The proud (racial name) of (the) (civilization name) shall wipe you from the face of this galaxy!”

Friends List:

Another interface will list every other user whom you’ve added to your “friends list”; these users are the only ones that can take part in a private game (see Section 3 for that particular game option). To add a player to your Friends List you can either type in their name and search the server database for all users, or you can see a list of users involved in your current game(s) and select one or more of them to add. You can also remove a friend from this list. Note that there won’t be any sort of notification to another user that they’ve been added to or removed from this list – it’s unnecessary as this isn’t a social networking site, and regardless, the only purpose of adding someone to this list is that you are allowing them to join one of your private games.

Leader Boards:

The final interface a player can get to is the leader boards; this will list the top players in the game and will be sort-able by a variety of means (most wins, most ties, game type, etc.). Note that – rather than implementing this as part of the (Flash/iPad) client, we will likely just open the user’s browser and redirect them to a website (since wiring this up dhtml should be much simpler/quicker to implement than inside the client, not to mention making it easier to expand upon based on player feedback/requests/etc).

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